

FALCON's Response to RFQ Weight base rejection system 2.5K

Option-2

December 21, 2023



www.falconautotech.com

Kind Attention –

Mr. Sandesh Mulik

M/s Flipkart

Offer Ref: F23-00535_OP2; Date: 21-12-2023

Subject – Techno -Commercial Offer of Weight base rejection system 2.5K PPH

Dear Team,

We are pleased to submit our Techno-commercial Offer in response to your RFP

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies.

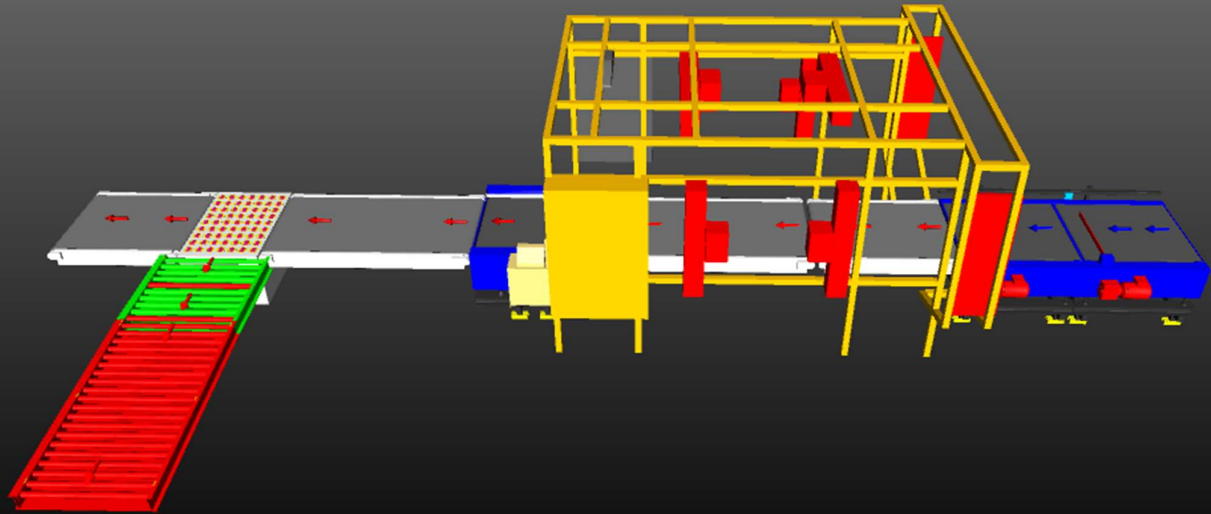
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFP for Weight base rejection system 2.5K PPH Throughput



Proposal Reference – F23-00535-OP2

**Falcon Autotech Private Limited
Plot No. 87, Sector Ecotech-1,
Extention-1, Greater Noida, Uttar
Pradesh 201308.**

**Contact – Sanyog Pratap Singh
Asst. Manager
Mob - +91 8750052591
Sanyog.singh@falconautoonline.com**

Table of Contents

1.	Glossary	6
2.	Company Profile	7
3.	Falcon's Experience and Achievements in Sortation Space Globally	11
4.	Reference Projects	13
4.1	Project 1- (CEP Client, India)	13
4.2	Project 2- (Client – E-Commerce, India)	14
4.3	Project 3- (Ecommerce Client, India)	15
4.4	Project 4- (CEP Client, UK)	16
4.5	Project 5- (CEP Client, Sydney)	17
5.	About Falcon	18
6.	Weight base rejection system	25
7.	Handled Parcel Spectrumions Minimum Maximum	25
7.1	Parcel size Conveyable on the entire system.	25
7.2	Parcels to be loaded on Conveyor/Sorter shall have the following characteristics:	26
7.3	Parcel not loadable on the system.....	26
7.4	Types of Parcels	27
8.	Proposed System Description	27
8.1	Objective	27
8.2	Process Flow	27
8.3	Main benefits of the proposed solution	28
9.	System Components	28
9.1	Spacer	28
9.2	Auto Barcode Scanning System (Top Side)	29
9.3	Weighing Conveyor	30
9.4	Swivel Wheel Sorting Unit – Falcon	31
9.5	PVC Belt Conveyor	32
9.6	Powered Roller Conveyor	33
9.7	Idler Roller Conveyor	34
10.	Proposed System Technical Details/BOM.....	35
10.1	Mechanical equipment.....	35
10.2	Electrical Equipment	35
10.3	Control System	35



11. Program Organisation 36

11.1 Delivery Schedule 36

12. Commercial Terms 36

12.1 Price Sheet 36

12.2 Payment Terms 36

13. Warranty Period 37

14. Exclusions 38

1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Parcels Per Hour
3	IT	Information Technology
4	BOQ	Bill Of Quantity
5	I/O	Input/ Output
6	PDP	Power Distribution Panel
7	PC	Personal Computer
8	UPS	Uninterrupted Power Supply

2. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



Operational since 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces



Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS



Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5
Product Lines

15+
Countries with Live Installations

1800+
Total Installed Systems Globally

1Million+
Sorts Per Hour

600+
Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

توتو

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Mohandeen (Dubai) Limited

News & Articles



FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.



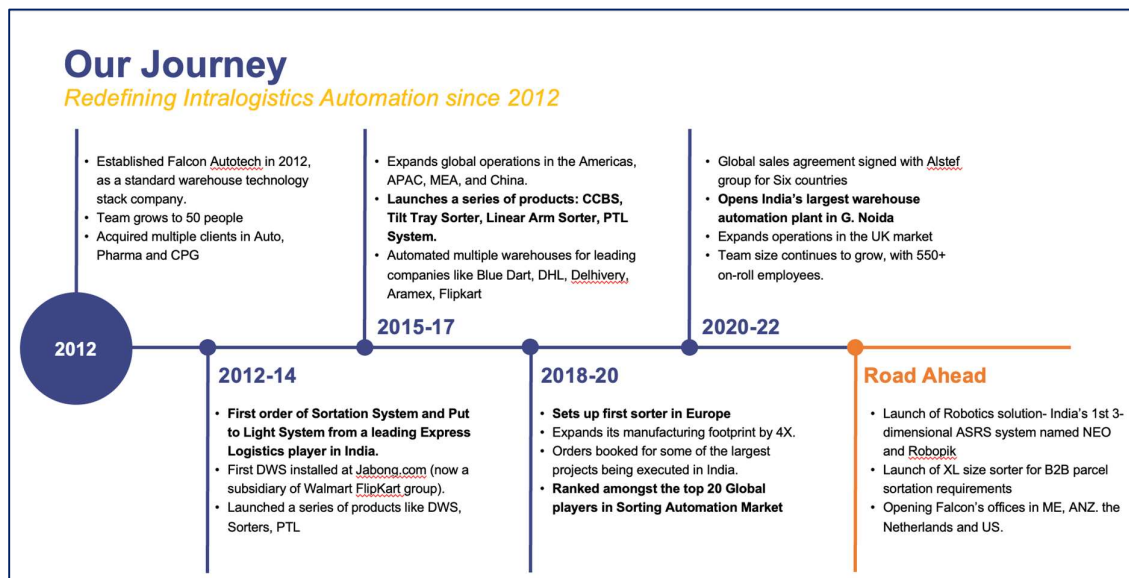
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- PopUp Sorter
- Sweep Sorter
- Pusher Sorter



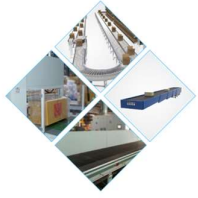
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

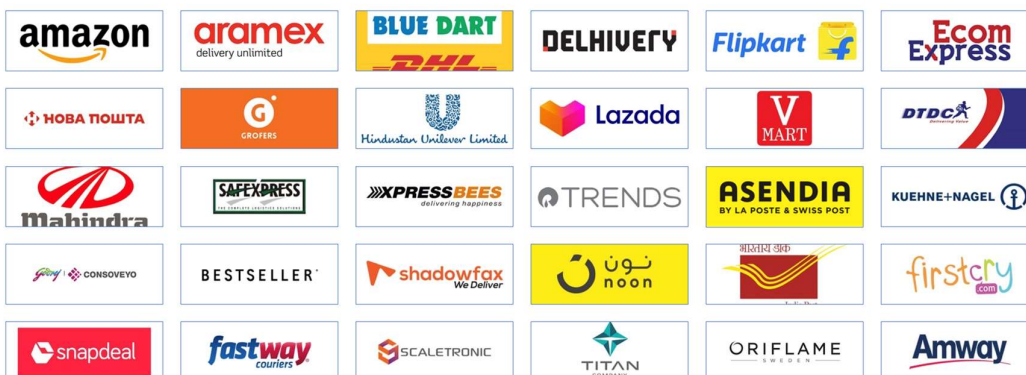


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

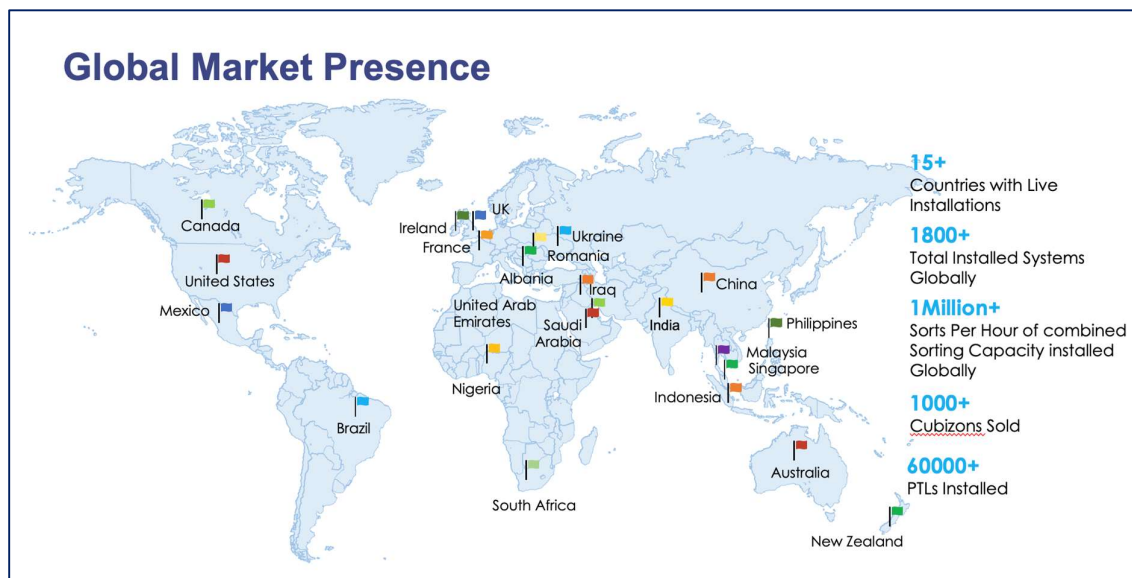
Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

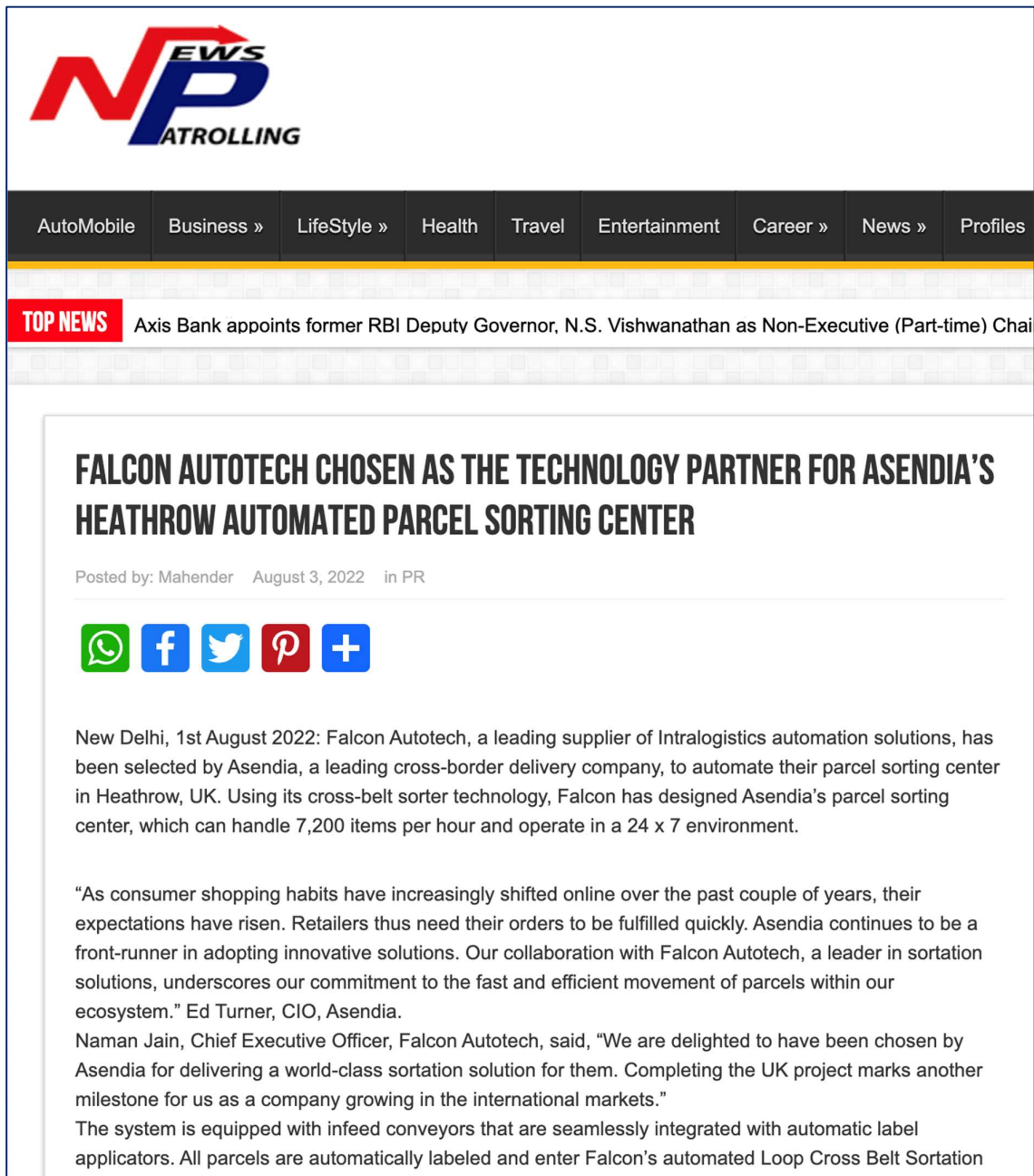


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



NP NEWS ATROLLING

AutoMobile Business » LifeStyle » Health Travel Entertainment Career » News » Profiles

TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

WhatsApp Facebook Twitter Pinterest +

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

4. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

4.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- | | |
|---|--------------------------------------|
| • 2 Sets of Double Decker CBS Sorters | • Modular Belt Conveyors |
| • Mezzanine Structures (one large mezz on to install a double decker Sorter). | • Spiral Chutes with Braking rollers |
| • Automated Singulators | • 5-Sided Scanning Tunnels |
| • Fully Automatic Inductions | • High speed weighing conveyors |
| • Semi-Automatic Inductions | • Direct Bagging Chutes |
| • Telescopic belt conveyors | • Put to Light Chutes |
| • PVC Belt Conveyors | • Volume Distribution systems |
| | • High Availability Server Systems |
| | • WCS |

Site Pictures -



4.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

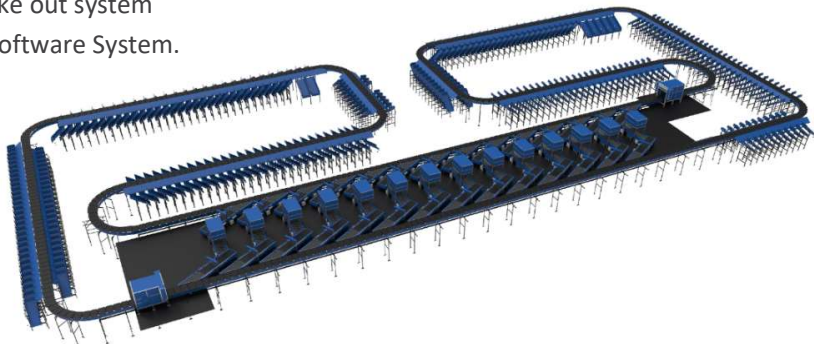
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



4.3 Project 3- (Ecommerce Client, India)

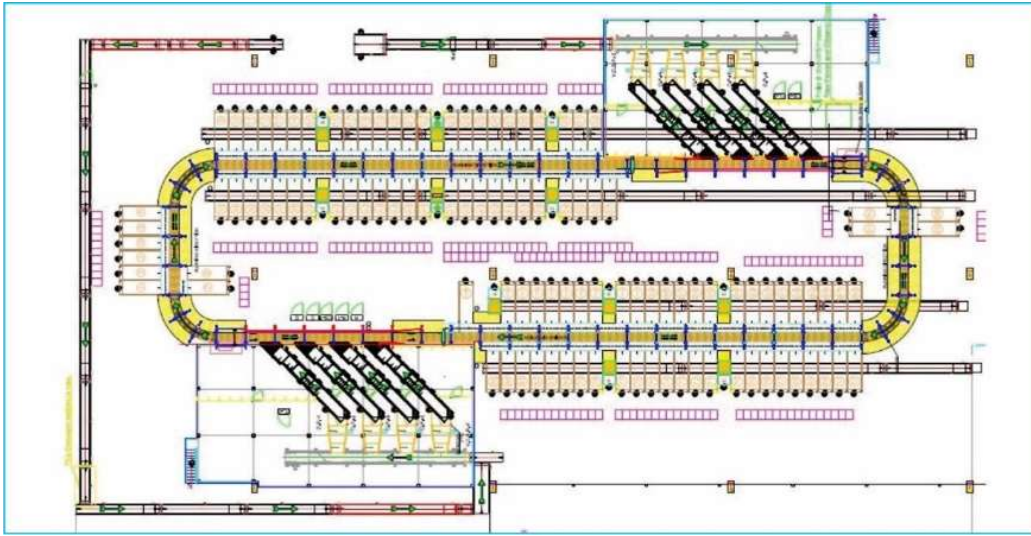
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



4.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



4.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

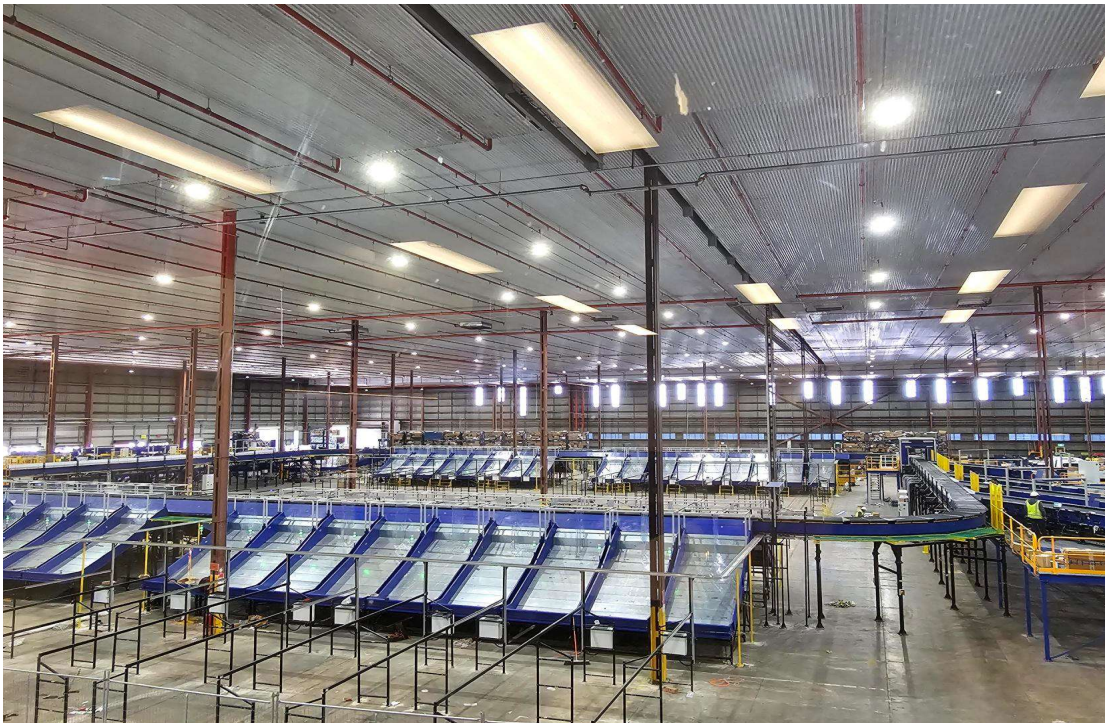
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

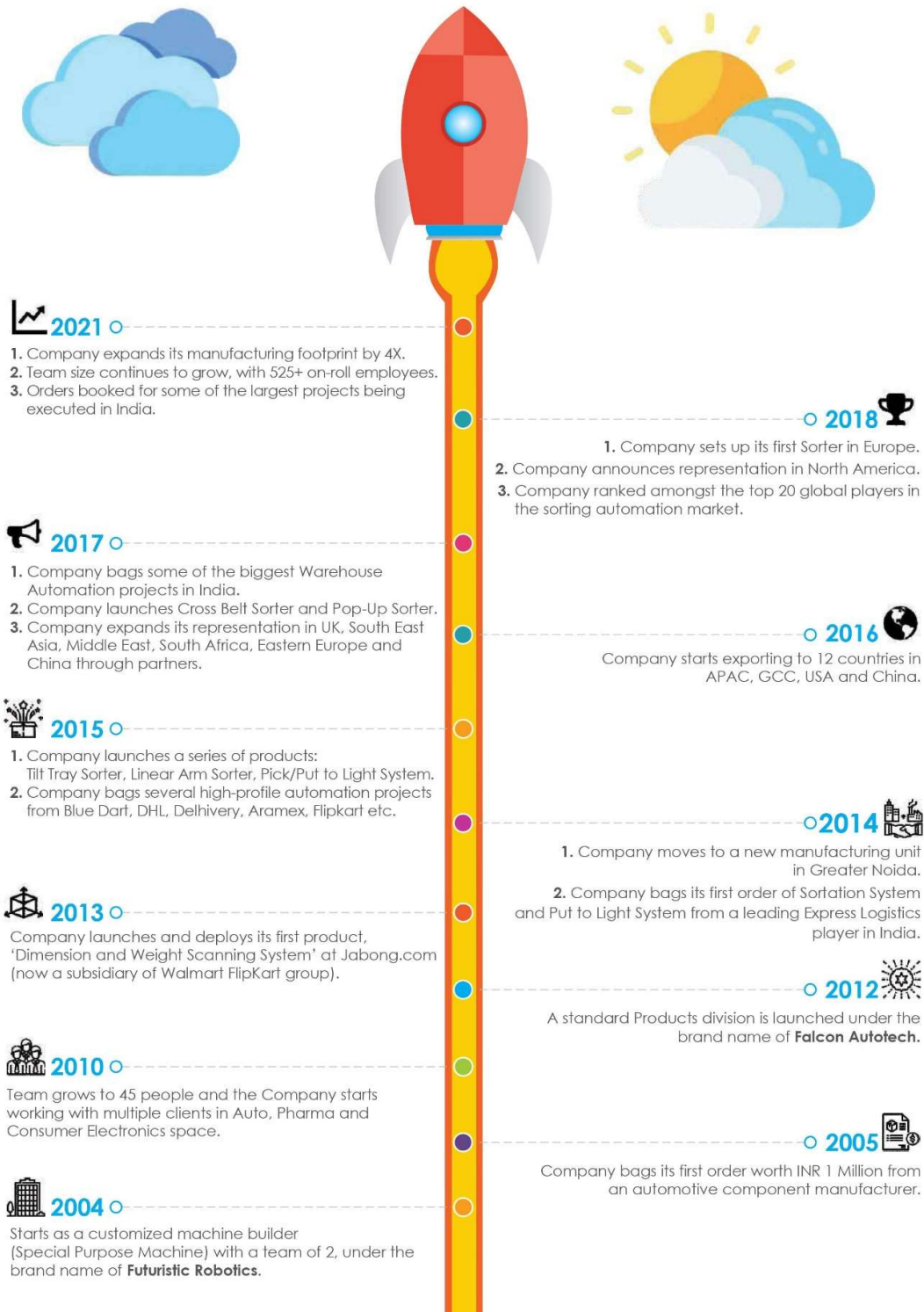
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



KEY STATISTICS

**04**Product Lines

**525+**On-Roll Employees

**290+**Engineering Team

**400000+ Sq Ft**Manufacturing Space

**20+**Tier A Clients

**15+**Countries with Live Installations

**1800+**Total Installed Systems Globally

**35**Representations in 4 Continents and 35 Countries

**1Million+**Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04

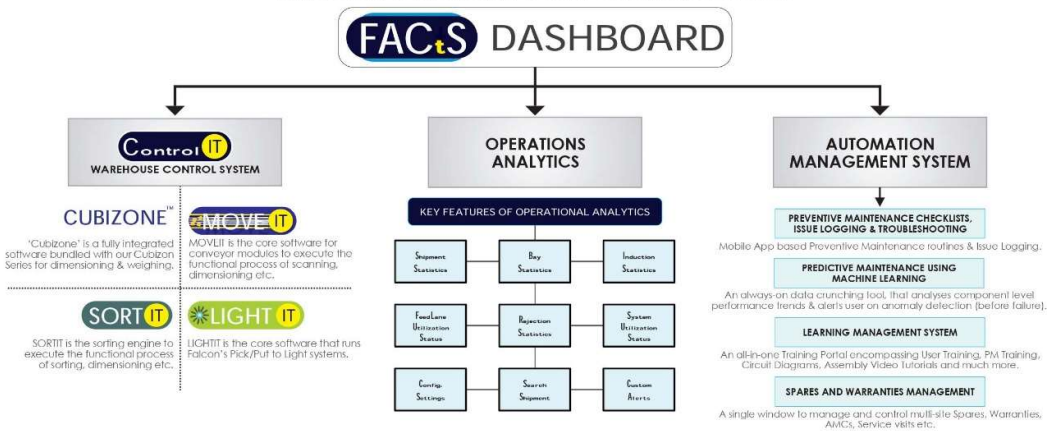
CONVEYOR AUTOMATION SOLUTIONS

BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor

05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



ADD-ON MODULES



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DUC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



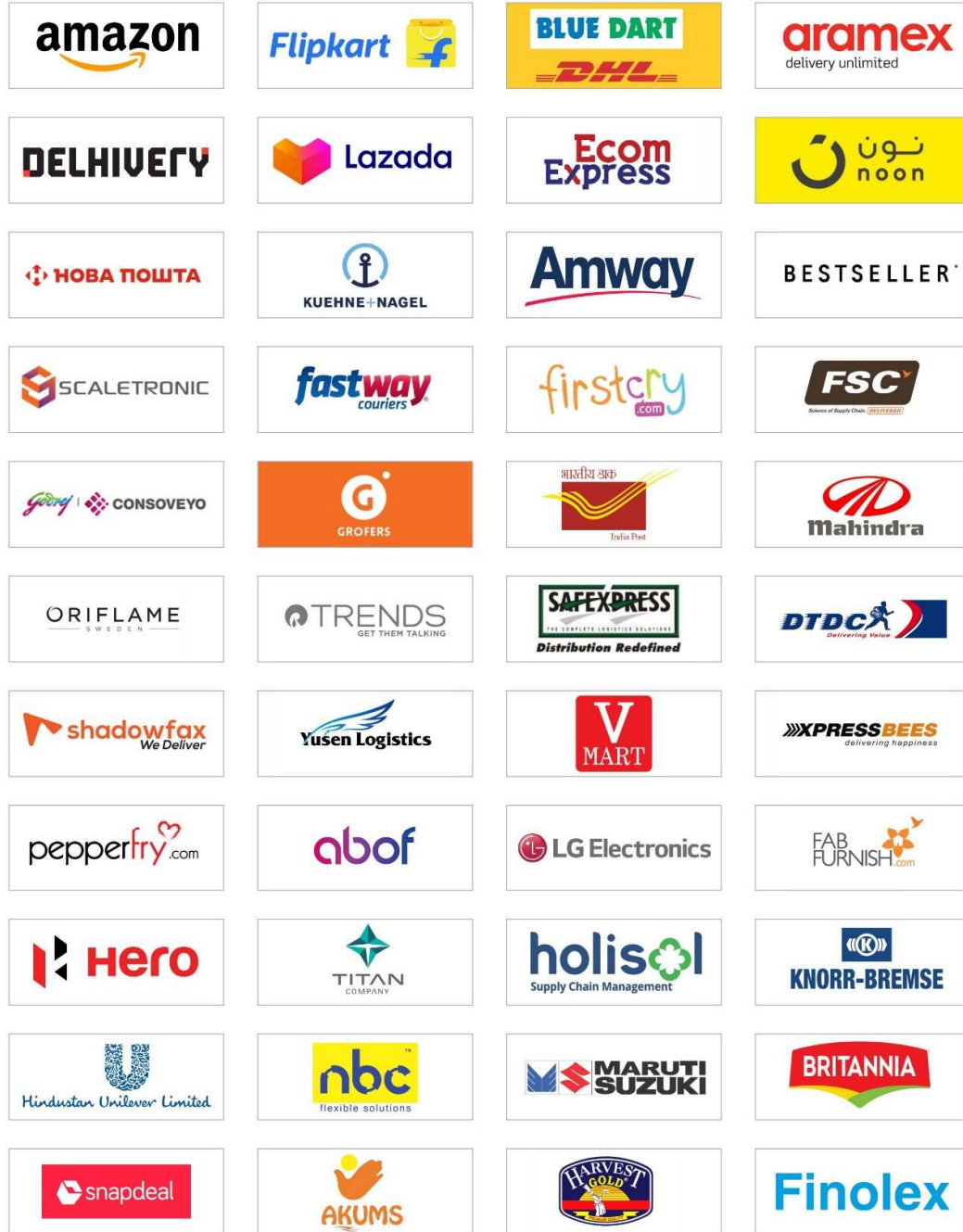
SPARE PARTS DISTRIBUTION



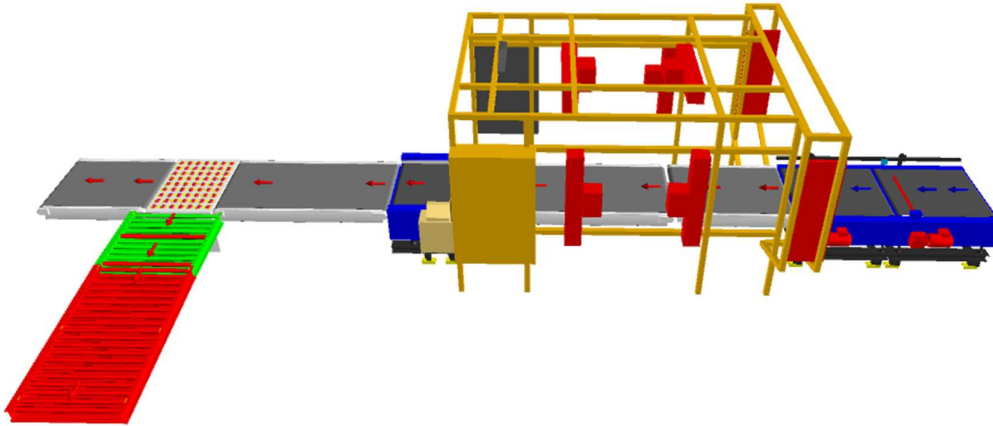
GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



6. Weight base rejection system



7. Handled Parcel Spectrum

7.1 Parcel size Conveyable on the entire system.

Specification	Unit	Value
Max Length	mm	800
Max Width	mm	800
Max Height	mm	600
Max Weight	Kg	35
Min length	mm	150
Min Width	mm	150
Min Height	mm	50
Min Weight	Kg	0.1
Weighing Accuracy	Kg	+/- 0.06 or 0.5 %, whichever is higher

7.2 Parcels to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of Item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behaviour of parcel cannot be guaranteed.
3. Parcels shall be perfectly and safely packed: protrusion or open surfaces are not allowed.
4. Plastic ropes shall be perfectly adherent to the surface of the package.
5. Items must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
6. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
7. Items shall not have slippery surfaces and must be able to withstand acceleration of the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
8. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
9. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
10. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

7.3 Parcel not loadable on the system

All products that are within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll on the sorting system, such as spherical or cylindrical items
2. Items that are packed in material that can damage the conveyors or the sorter.
3. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors
4. Items that have been classified as dangerous.
5. Wet items.
6. Items with anti-slip treatment.
7. Items with protruding parts.
8. Inadequately packed items that could be damaged during automatic transportation.
9. Electrostatically loaded items.
10. Loose parts on Items such as adhesive tape, stickers, slips of paper, straps, wrap foil etc.
11. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint.
12. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system.

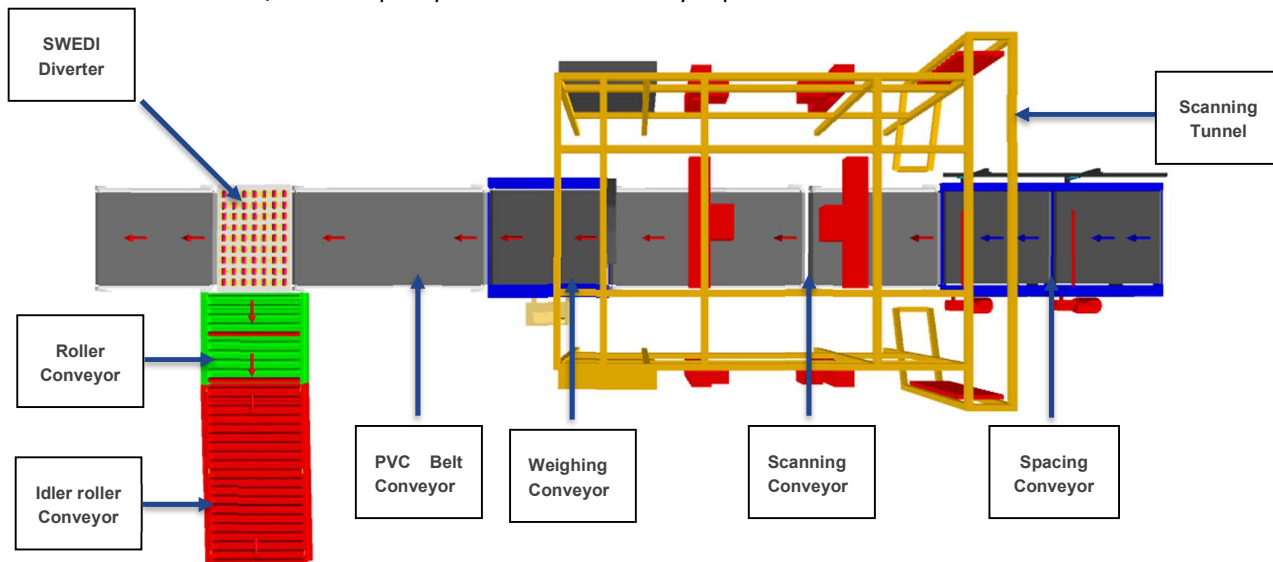
7.4 Types of Parcels

Parcel Type	Description
Corrugated Box/Cartons	Min: L-150 X W-150 X H-50 MM
Shipments	Max: L-800 X W-800 X H-600 MM

8. Proposed System Description

8.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the fully auto Weight base rejection system to handle cartons/boxes as per specifications shared by Flipkart.



Proposed System Layout

8.2 Process Flow

The shipments will be loaded on the existing conveyors placed before the spacing units. The shipment passes through spacing conveyors creating enough space between parcels and enters the automatic barcode scanning tunnel (Six Side barcode scanning) where the barcode data is read, the weight is measured and stored in the system. In case of barcode not being read or overweight the parcel will be rejected by the SWEDI divert onto the rejection roller conveyor. In case of a successful data capture & allowable weight the parcels shall head straight onto the output Conveyor.

8.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Low occupancy of floor space in the building

9. System Components

9.1 Spacer

It is a Dual set of high-speed conveyors, using to create appropriate space between parcels/boxes.

Some Salient Features of Falcon's spacing conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



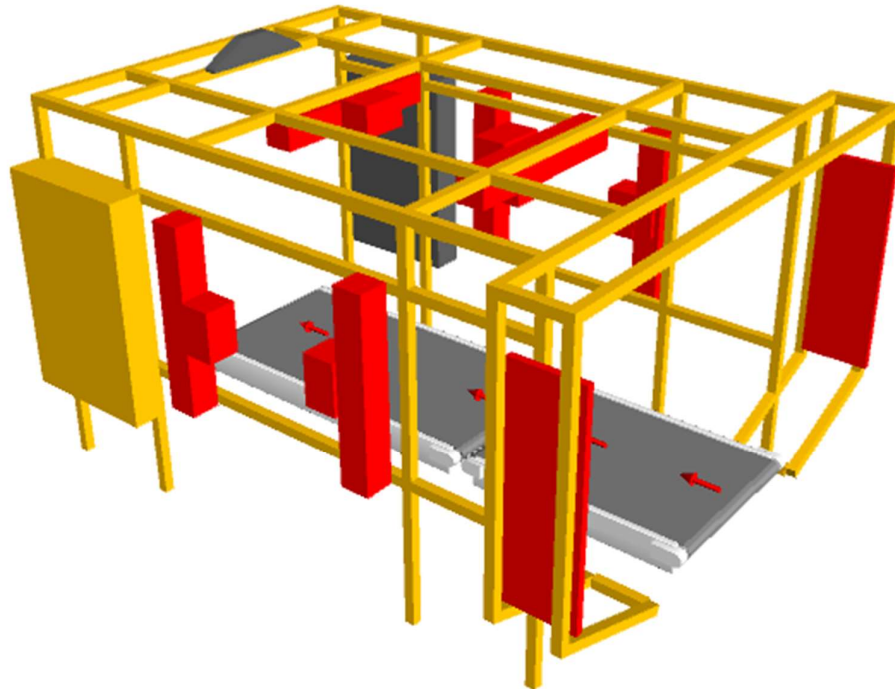
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Type of Motor	Type	AC Geared Motor
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type

9.2 Auto Barcode Scanning System (Top Side)

System is integrated with automated Camera technology for scanning 1D/2D Barcodes.

Some Salient Features of Falcon's Scanning System:

- High Speed Scanning
- High Accuracy Rates
- Effortless Operation
- High safety standards.



**Note : Barcode data of product should be locally available for on time retrieval of weigh data against each barcode.*

**Note : In case of six Side scanning, skewness of more than +/- 10 deg may result in no data capture case.*

9.3 Weighing Conveyor

Falcon's weighing conveyor is used for inline weighing of parcels.

Some Salient Features of Falcon's weighing conveyor:

- High Speed Weighing
- Minimal maintenance
- High safety standards.
- High accuracy



Specification	UOM	Remark
Manufacturer Name	Name	Mettler Toledo/Bizerba
Carton Size	Mm	Min: 150 x 150 x 50 Max: 800 x 800 x 600
Minimum Weight	Kg	0.1
Maximum Weight	Kg	35
Weighing Accuracy	Kg	+/- 0.06 or 0.5% of product weight, whichever is higher
Weighing Resolution	Kg	0.001

9.4 Swivel Wheel Sorting Unit – Falcon

Falcon's Swivel wheel divert is used in system for the smooth transfer of products.

Some Salient Features of Falcon's Swivel Wheel Divert:

- Low Noise
- Maximum uptime.
- Electric based rotation of wheels.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120
No. of divert sections	Yes/No	1 or 2
Max. product weight (kg/m2)	kg/m2	25
Throughput	PPH	2500
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

9.5PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



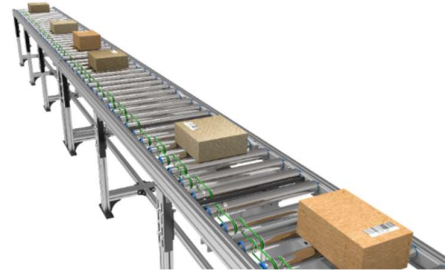
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Pitch on return side (Bottom)	mm	1200
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns

9.6 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	1000
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity/unit length of Conveyor	kg/m	50
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	75/100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

9.7 Idler Roller Conveyor

Falcon's Roller conveyors require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's idler roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	1000
Conveyor Height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Pitch of Rollers	mm	75
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes

10. Proposed System Technical Details/BOM

10.1 Mechanical equipment

S.No.	Qty.	Description	Value
1	1	Weight base rejection system	
		Spacing Conveyor	2 Set
		Barcode Scanning Conveyor	2 Set
		Weighing Conveyor	1 Set
		Automatic Barcode Scanner (Six Side)	1 Set
		PVC Belt conveyor	2 Modules (Length-3.2 m)
		SWEDI Divert unit	1 Set
		Powered Roller conveyor	1 Module (Length-1 m)
		Gravity Roller conveyor	1 Module (Length-2 m)

10.2 Electrical Equipment

Electricals			
FC1	1	Electrical Cabling and Wiring	
		Main Power Distribution panel	
		Main Control Panel	
		Network Switches	
		Field Cabling	Included

10.3 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls	
		Weigh Scale Control, Barcode Scanner Controls, Control IT Software Package	Included

11. Program Organisation

11.1 Delivery Schedule

Dispatch within 14-15 weeks from the date of signed PO, advance and DAP whichever is later.

12. Commercial Terms

12.1 Price Sheet

S. No	Component	Set	Price (INR)
1	SWEDI Diverter	1	₹ 10,07,232
2	Scanning & Weighing System	1	₹ 95,57,082
3	Conveyor Automation	1	₹ 23,23,650
4	Controls	1	₹ 8,00,000
5	Toolkit	1	₹ 36,000
6	Software Packages	1	₹ 6,00,000
7	Installation & Commissioning	1	₹ 6,86,198
8	Packaging & Forwarding	1	₹ 2,74,479
9	Project Management and Engineering Charges	1	₹ 3,82,116
Grand Total (INR)			₹ 1,56,66,758

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual
- Taxes, Customs, Duties extra at actuals
- Including Packing and Forwarding
- Installation and Commissioning: Mentioned above.
- Price is valid for 60 days from the date of proposal

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

12.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 30% along with 100% Taxes prior to dispatch.

Stage 3: 20% Within 30 days of Commissioning.

13. Warranty Period

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

14.Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Low Spec Server PC
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Server PC is not included in this proposal
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.